



Inferring Security-relevant Properties from Java Documentation

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Abstract

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Contents

1	Introduction	5
2	Background Knowledge	7
3	Literature Review	9
4	Milestone Task 1	11
5	Milestone Task 2	13
6	Conclusion	15
A	Appendix	17

Chapter 1

Introduction

Static application security testing (SAST) applies static program analysis to the security of software projects. Static analysis is performed on the source code, compiled byte-code, or some other representation of the source code [1]. Popular SAST tools include: SonarQube, SpotBugs, Snyk Code, and Facebook’s Infer. SAST tools have become widely used at all stages of the software development life-cycle. Furthermore, they have been demonstrated to be highly effective in identifying defects in code compared to other methods, such as manual inspection [2].

Unfortunately, SAST tools are not perfect. From Rice’s Theorem, it is known that any non-trivial property about programs is reducible to the halting problem, i.e., undecidable [3]. That is, a perfect analysis is not always possible, and approximations must be made in order for the analysis to terminate, resulting in an imperfect output. False positives (where safe code is incorrectly labelled as a defect) and false negatives (where the analysis misses a genuine fault) [4] are common in all static analysis tools. False positives are not ideal as manual inspection is required to isolate genuine defects, leading to an increased time and labour cost. And, false negatives of course mean that genuine software faults are left undetected by the analysis. While solving the halting problem may be out of scope for an undergraduate thesis, there are still additional methodologies for improving existing static analyses that do not involve upsetting Mr. Turing.

To reduce the rate of false negatives, manual configuration of the SAST tools for specific software projects is often required [5]. For example, to perform an effective taint analysis, a custom configuration of sources, sinks and sanitisers for a particular project is often required. These taint analysis tools are not sophisticated enough to implicitly categorise methods as a source, sink, sanitiser — they require sensitive methods to be pre-labelled to conduct the analysis. Generally, these tools come with a default configuration containing a list of sensitive methods in popular APIs. For

example, the `getParameter()` method as a part of the `ServletRequest` interface would likely be listed by all taint analysis tools as a source. Of course, if a project is using a niche library or code developed in-house, the taint analysis will not be able to incorporate those data-points into its analysis, leading to sub-optimal output. To improve the taint analysis, one can always identify the sensitive methods manually and add these to the taint analysis' configuration. However, creating this manual configuration is often a time-consuming and resource-intensive process. As such, methodologies to automatically label security sensitive methods may enable existing taint analysis tools to be configured more efficiently and effectively.

Deep learning, a subtopic within the broader machine learning, has recently increased in popularity. What was once a niche topic reserved for academics with access to computational resources far out of reach by most has become arguably one of the most influential technologies of the modern age. Deep learning has fundamentally changed our everyday life, with applications ranging from targeted advertising to identifying COVID-19 cases from lung CT scans [6]. Natural-language processing (NLP) is one often overlooked application of deep learning, but it has seen tremendous growth in recent years.

NLP has existed in one form or another for decades. In the early days of NLP, work was painstakingly done using handwritten rules, before moving to probabilistic methods in the 1990s, and then to deep learning in the 2010s [7]. With tech conglomerates such as Google and Microsoft publicly releasing large pre-trained models, NLP has never been more accessible. While machine translation and speech recognition are among the most popular applications of NLP, one application that has not been explored thoroughly in the literature is software security. While being an informal description of code, Javadoc (i.e., natural language descriptions of code) can contain information relevant to software security. Using CodeBERT, a powerful pre-trained model by Microsoft, that understand the semantic connection between code and natural language [8], it may be possible to extract security-relevant properties from Javadocs and use this information to configure existing SAST tools so they may operate more effectively.

In this thesis, we will investigate X different applications of deep-learning analysis of Javadoc as it pertains to security, and...

Chapter 2

Background Knowledge

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Chapter 3

Literature Review

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Chapter 4

Milestone Task 1

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Chapter 5

Milestone Task 2

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Chapter 6

Conclusion

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Appendix A

Appendix

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